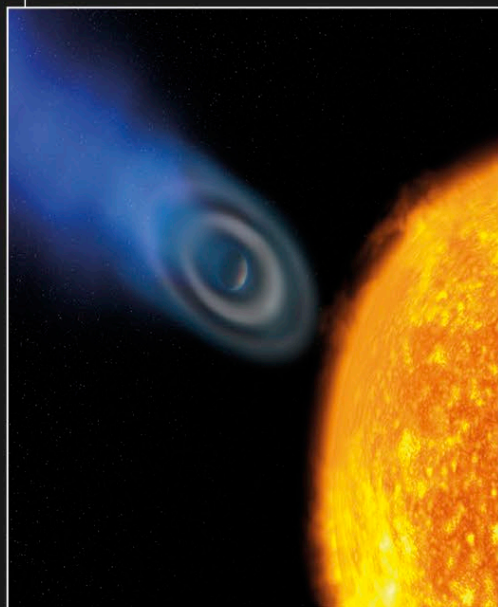


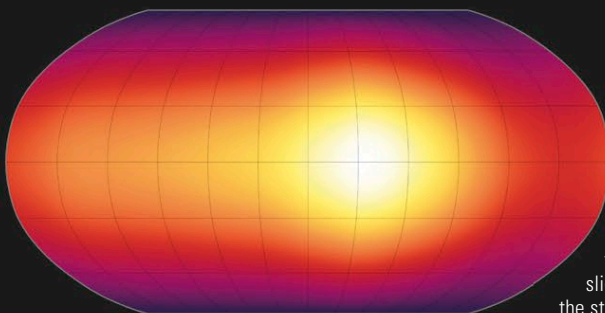
GAS GIANTS

The first exoplanets to be discovered around Sun-like stars were all giants with masses from about that of Neptune to much larger than Jupiter, with short-period orbits. While this reflects the fact that planets of this type are easiest to find with the radial velocity method (and many giants at greater distances have since been discovered), the existence of these so-called “hot Jupiters” in large numbers was a surprise. According to current theories, such giant planets must have formed farther out from their stars and spiraled inward to their current orbits. Temperatures in the atmospheres of hot Jupiters frequently exceed 2,000°F (1,100°C), causing them to slowly evaporate into space. More recently, several possible “Chthonian” planets—the bare solid cores of former hot Jupiters—have also been identified.



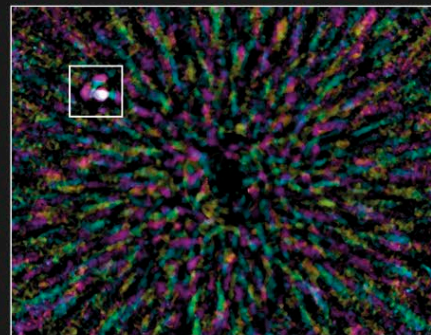
ESCAPING ATMOSPHERE

The “hot Jupiter”-type exoplanet HD 209458b orbits close-in on its parent star (seen here in an artist's impression). In 2003–2004, astronomers discovered an extended ellipsoidal envelope containing hydrogen and other gases evaporating from the planet. It is thought this type of atmosphere loss may be common to all “hot Jupiters.” When all their volatile materials have evaporated away, an exposed rocky or metallic core may survive as a “Chthonian planet.”



KAPPA ANDROMEDAE B

This rare view shows the massive “super-Jupiter” planet orbiting Kappa Andromedae, a hot young star some 170 light-years from Earth (whose direct light is blocked in this image). Kappa Andromedae b has 12.8 times the mass of Jupiter, putting it right on the borderline between a giant planet and a brown dwarf “failed star.”



51 PEGASI B

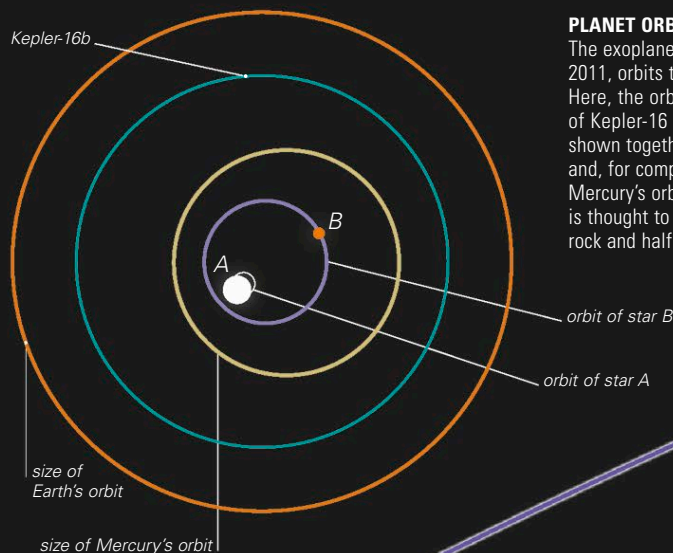
This artist's impression shows the first exoplanet discovered around a Sun-like star—initially nicknamed Bellerophon but now formally named Dimidium. Lying some 50 light-years from Earth, 51 Pegasi B has a mass at least half of Jupiter's, and orbits its star at an average 1/20th of the Earth–Sun distance, making it the prototype of the “hot Jupiter” class of giant planets.

EXOPLANET TEMPERATURES MAP

This map produced by the Spitzer Space Telescope shows temperature variation across the surface of the “hot Jupiter”-type exoplanet HD 189733b. One side of the planet always faces its parent star. The hottest area is slightly displaced from the point on the planet exactly facing the star—evidence that fierce winds operate in its atmosphere.

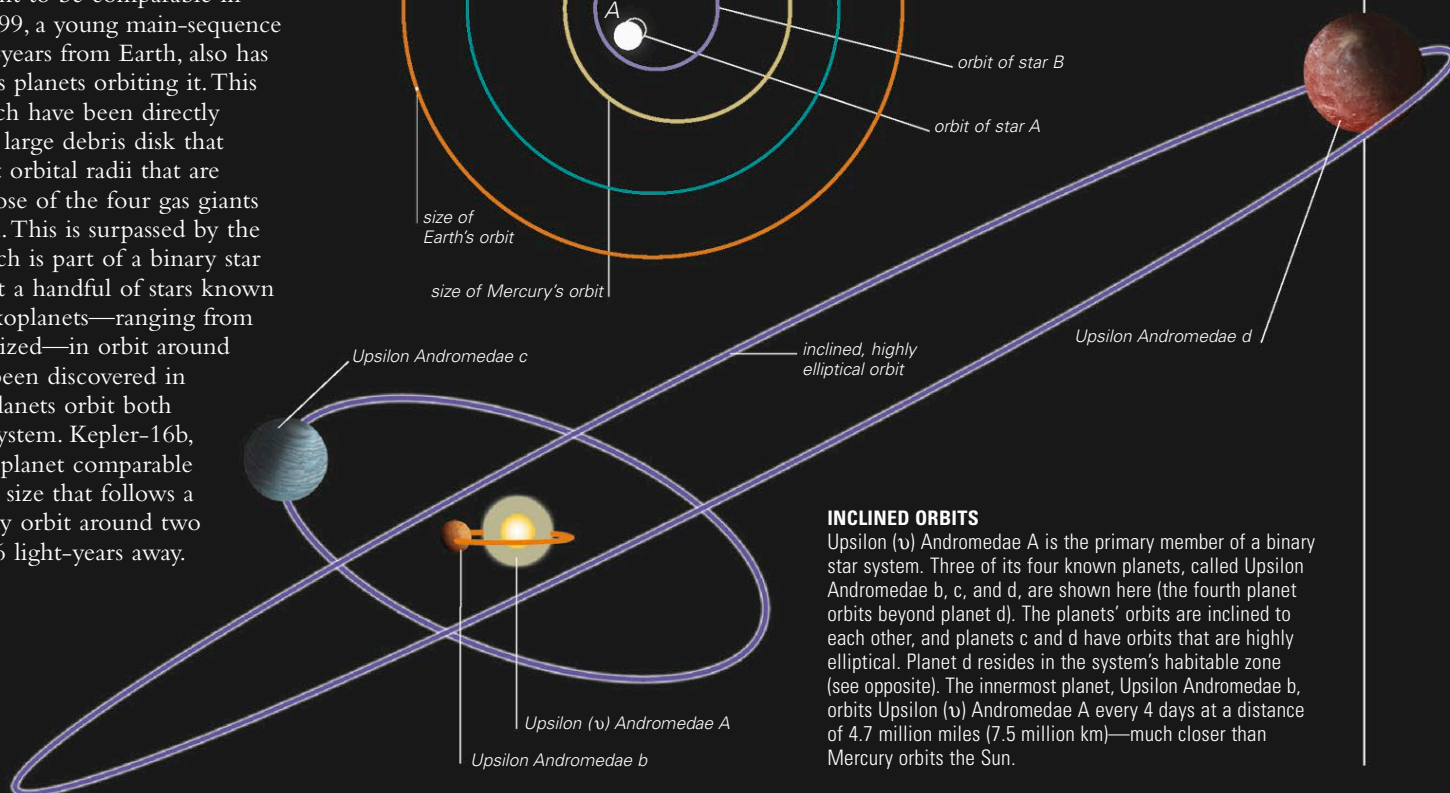
PLANETARY SYSTEMS

Multiple exoplanets have been observed orbiting a number of relatively close stars. The first system of this type to be identified, in 1999, was found orbiting Upsilon (υ) Andromedae A, a Sun-like star located approximately 44 light-years away. The system is now known to include at least four planets, all thought to be comparable in size to Jupiter. HR 8799, a young main-sequence star located 129 light-years from Earth, also has at least four high-mass planets orbiting it. This quartet of giants, which have been directly imaged, orbit inside a large debris disk that surrounds the star—at orbital radii that are two to three times those of the four gas giants orbiting our own Sun. This is surpassed by the star 55 Cancri A, which is part of a binary star system and one of just a handful of stars known to have at least five exoplanets—ranging from Neptune- to Jupiter-sized—in orbit around it. Systems have also been discovered in which one or more planets orbit both stars in a binary star system. Kepler-16b, for example, is an exoplanet comparable to Saturn in mass and size that follows a nearly circular 229-day orbit around two stars located some 196 light-years away.



PLANET ORBITING A BINARY STAR

The exoplanet Kepler-16b, discovered in 2011, orbits the binary star system Kepler-16. Here, the orbits of the two components of Kepler-16 (labeled stars A and B) are shown together with the orbit of Kepler-16b and, for comparison, the size of Earth's and Mercury's orbits around the Sun. Kepler-16b is thought to be made up of about half rock and half gas.



INCLINED ORBITS

Upsilon (υ) Andromedae A is the primary member of a binary star system. Three of its four known planets, called Upsilon Andromedae b, c, and d, are shown here (the fourth planet orbits beyond planet d). The planets' orbits are inclined to each other, and planets c and d have orbits that are highly elliptical. Planet d resides in the system's habitable zone (see opposite). The innermost planet, Upsilon Andromedae b, orbits Upsilon (υ) Andromedae A every 4 days at a distance of 4.7 million miles (7.5 million km)—much closer than Mercury orbits the Sun.